Performance of Large Language Models on travel medicine certificate multiple choice questions

Angelo D'Ambrosio†[1], Francesco Baglivo†[2], Luigi De Angelis†[2], Federico Tecchio[2], Caterina Rizzo[2]

Introduction

Large language models (LLMs) now approach **professional-level performance** on medical MCQs.

Travel medicine is dynamic and under-studied: region-specific risks and evolving recommendations.

We benchmarked diverse models and prompting strategies on single-best-answer questions.

Methods

Questions: 42 travel-medicine items (single correct option out of four).

Models: A mix of frontier, open-source, and local quantised models.

Prompt strategies: "cold" (direct answer only), "free" (brief rationale + answer), and "reasoning" (CoT).

Replications: 5 per (item, model, prompt) cell to assess variability.

Metrics: Accuracy, parsability (output format adherence), consistency (replication stability).

Bayesian modelling: Hierarchical models to model model impact adjusted for items and prompt strategies.

Results

Reasoning models (e.g., o3, Gemini 2.5 Pro) led the benchmark, especially if equipped with **internet search capabilities** (e.g., Sonar Reasoning).

Performance also **dropped with parameter count**.

Local low-parameter count models performed worst.

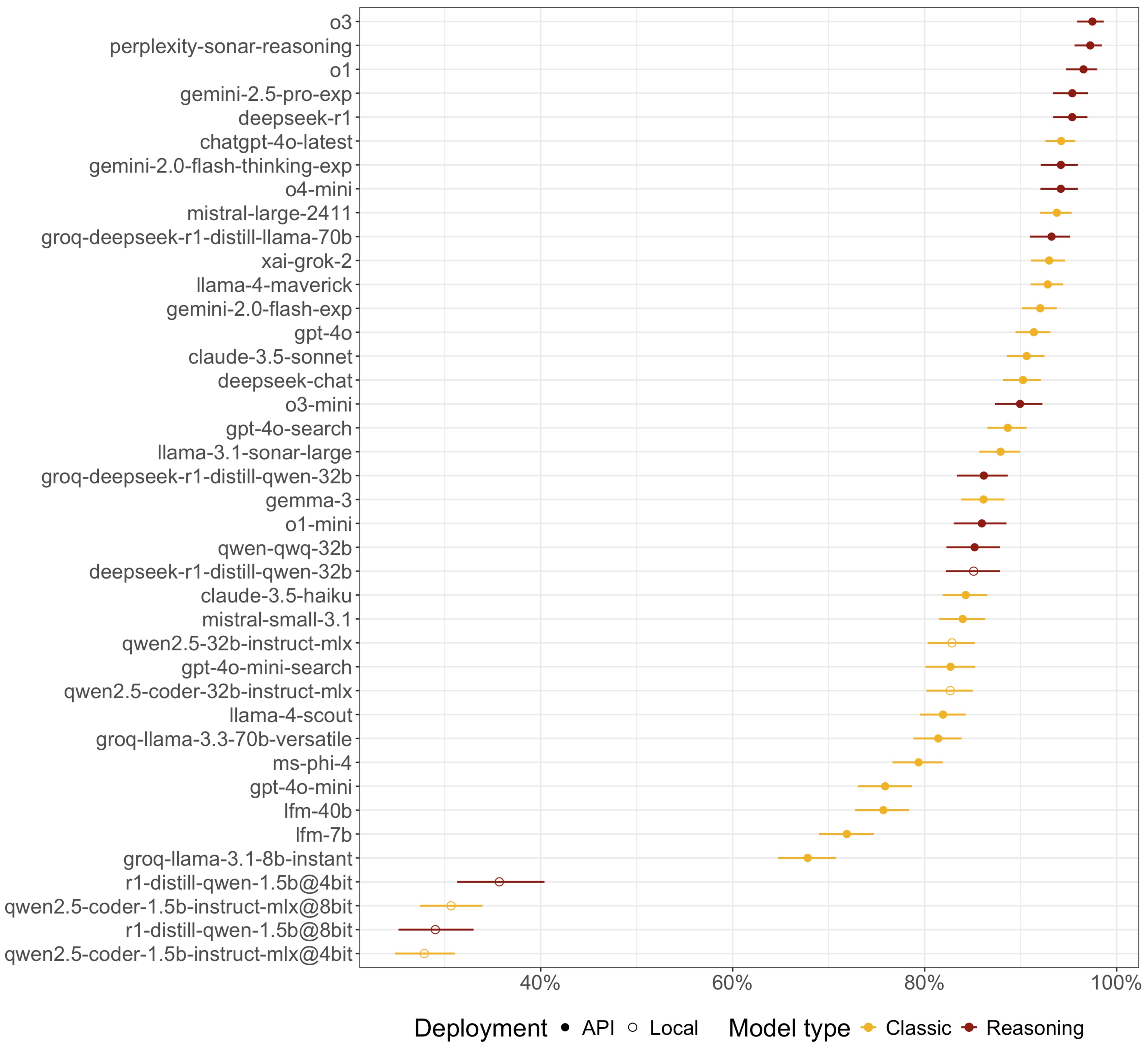
Prompting strategies had modest impact at the top end; model class mattered more.

o3 consistently failed only on 1 question, but the answers **nevertheless followed guidelines**.

Questions	Models	Replications	Best Accuracy
42	40	5	97.5% (o3)

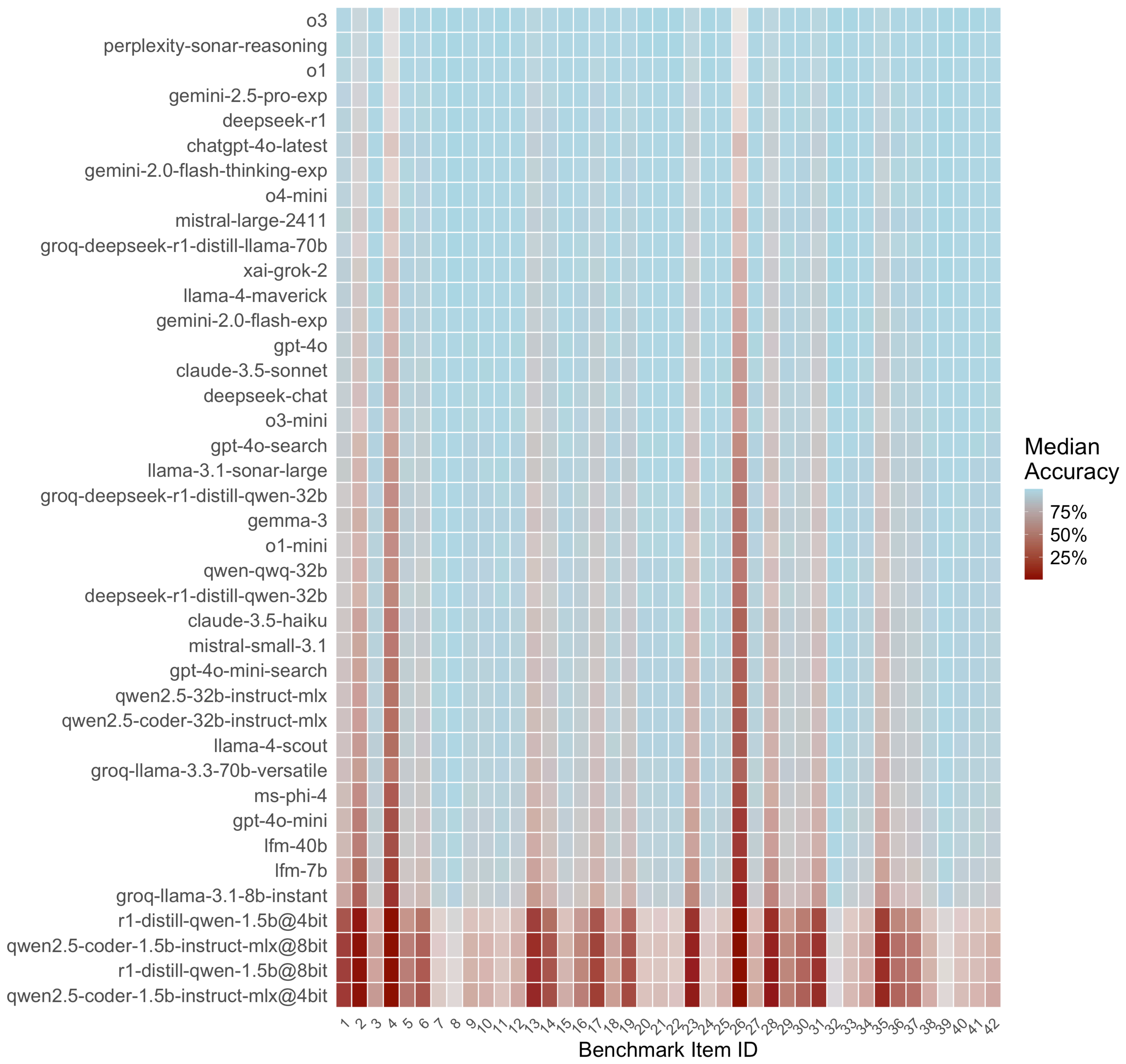
Accuracy by model

Posterior accuracy marginalised across items and replications.
Points: posterior median; bars: 95% CrI.



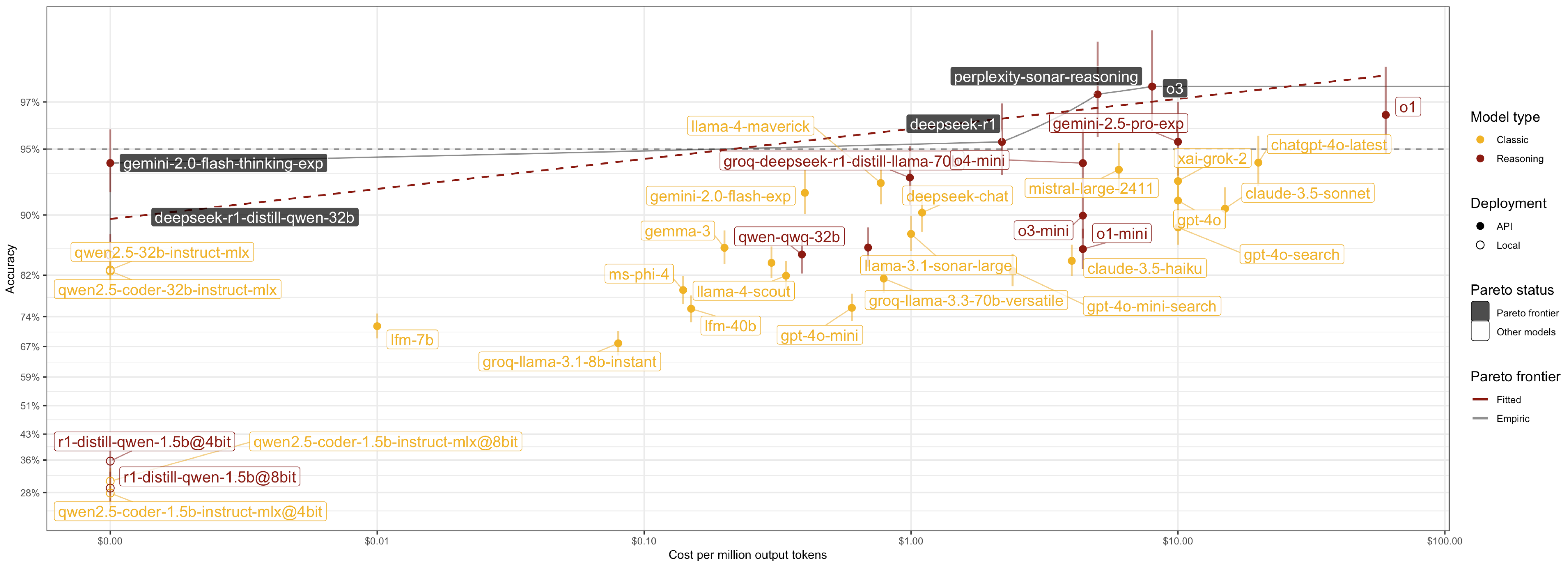
Item-level accuracy across models

Color indicates median accuracy; transparency reflects certainty (1 - CrI width).



Pareto frontier: Model accuracy vs. cost efficiency

Optimal models balance high accuracy with low operational costs.
Points = individual models; Lines = Pareto-efficient frontiers



Discussion

- Frontier models now reach ~95–98% accuracy on this benchmark.
- Search-augmented models can match high-end performance at lower cost.
- Local models are cost-efficient but currently less accurate on complex items.

Conclusions

- LLMs can provide clinically meaningful support in travel-medicine scenarios.
- Ongoing auditing and transparent reporting remain essential for deployment.

Contacts: angelo.dambrosio@ecdc.europa.eu; luigi.deangelis@unipi.it; francesco.baglivo@unipi.it

Affiliations: [1] ECDC; [2] University of Pisa

Contributions: † Denotes equal contribution